

Employment

Jun 2026 - present **Postdoctoral Scholar**, *Oregon State University*, Corvallis, OR
Mentor: A. Saenz

Education

2020 - 2026 **PhD in Mathematics**, *University of Virginia*, Charlottesville, VA
Advisor: L. Petrov

2015 - 2020 **BA in Physics**, *Lomonosov Moscow State University*, Moscow, Russia, 4.75 out of 5
Advisor: G. V. Koval
BA thesis: **"Exact solution of a spin chain in quenched disorder"**

Awards and grants

2026 **All-University Graduate Teaching Award**, *University of Virginia*
University-wide award for excellence in teaching by graduate students

2026 **ICM 2026 Travel Grant (\$2,400)**, *American Mathematical Society*, funded by the NSF
Travel support to attend the International Congress of Mathematicians, Philadelphia, PA

Keywords

integrable probability, statistical mechanics, interlacing particle systems
vertex models, symmetric functions, representation theory, algebraic combinatorics

Publications

2026 Mikhail Tikhonov. **Random permutations from q -Demazure products**. Submitted to *Electronic Communications in Probability*. arXiv: 2604.06532 [math.PR].

2023 Leonid Petrov and Mikhail Tikhonov. **Asymptotics of noncolliding q -exchangeable random walks**. In: *Journal of Physics A: Mathematical and Theoretical* 56. arXiv: 2303.02380 [math.PR].

2021 I. Mursenkova, M. Timokhin, M. Tikhonov, A. Militsina, and A. Kuznetsov. **Digital processing of shadowgraph images taking into account the diffraction of light at a shock front**. In: *Journal of Physics: Conference Series* 2127 Issue 1.

2020 Leonid Petrov and Mikhail Tikhonov. **Parameter symmetry in perturbed GUE corners process and reflected drifted Brownian motions**. In: *Journal of Statistical Physics* 181. arXiv: 1912.08671 [math.PR].

M. Yu. Timokhin, M. Tikhonov, I. V. Mursenkova, and I. A. Znamenskaya. **Shock-wave thickness influence to the light diffraction on a plane shock wave**. In: *Physics of Fluids* 32 Issue 11.

2019 Maksim Timokhin and Mikhail Tikhonov. **Numerical simulations of micro-channel devices with Lattice Boltzmann method**. In: *AIP Conference Proceedings* 2132 Issue 1.

Talks, posters

Jul 2026 **Random permutations from q -Demazure products**, *West Coast Coin Flippers Union, ICM Satellite Edition*, *University of Delaware*, Newark, DE

May 2026 **Random permutations from q -Demazure products**, *Oregon State University Probability and Data Science Seminar*, Corvallis, OR

- Jan 2026 **Random Permutations from Permutons from Stochastic Demazure Product**, *JMM Special Session on Random Tilings, Random Permutations, and Particle Systems*, Washington, DC
- Dec 2025 **Random Permutations from q -Demazure product**, *Geometry, Statistical Mechanics, and Integrability Reunion conference*, Lake Arrowhead, CA
- Aug 2025 **Writing Effective Letters of Recommendation**, *Annual Teaching Workshop at University of Virginia*, Charlottesville, VA
- Oct 2024 **Noncolliding q -exchangeable random walks**, *Ohio State University Probability Seminar*, Columbus, OH
- Apr 2024 **Asymptotics of Noncolliding q -exchangeable random walks and lozenge tilings**, *Southern California Discrete Mathematics Symposium at UCLA*, Los Angeles, CA
- Nov 2023 **Noncolliding q -exchangeable random walks and lozenge tilings**, *UVA Probability Seminar*, Charlottesville, VA
- Jul 2023 **Noncolliding q -exchangeable random walks**, *Probability and Algebra: New Expressions In Mathematics*, College Station, TX
- Sep 2022 **Markov jumps for GUE corner process**, *Graduate Student Probability Conference at University of Wisconsin - Madison*, Madison, WI
- Apr 2022 **Parameter symmetry in the perturbed GUE corners eigenvalues**, *Graduate Student Seminar at UVA*, Charlottesville, VA
- Nov 2021 **Markov jumps for GUE corner process**, *Integrable Probability and Related Fields from a Safe Distance*, online talk
- Feb 2021 **Permuting the GUE corners eigenvalues**, *UVA Probability Seminar*, online talk
- Dec 2020 **Parameter symmetry in perturbed GUE corners process**, *XVI Brunel – Bielefeld Workshop on RMT*, London, UK, online poster talk
- Jul 2018 **Numerical Simulations of Micro-Channel Devices with Lattice Boltzmann Method**, *31st International Symposium on Rarefied Gas Dynamics*, Glasgow, UK, poster
- Apr 2018 **Exact solution of a spin chain in quenched disorder**, *Lomonosov Conference*, Moscow, Russia

Extended Visits and appointments

- Mar - Jun 2024 **Core Participant**, *Long program "Geometry, Statistical Mechanics, and Integrability" at IPAM*
- Jan 2020 **Visitor**, *University of Virginia*, Project: "Parameter symmetry in perturbed GUE corners process", Supervisor: L. Petrov
- Spring 2018 **Research Intern**, *mathCCES RWTH*, Project: "Nonlinear problems based on Boltzmann equation", Supervisor: M. Torrilhon

Summer schools, conferences, short visits

- Jul 2026 **International Congress of Mathematicians**, *Philadelphia, PA*, AMS ICM 2026 Travel Grant
- Jul 2024 **Virginia Integrable Probability Summer School 2024**, *University of Virginia*
- Jun 2022 **Random Matrix Theory Summer School**, *University of Michigan*
- Jun 2022 **Summer school on Free Probability, Random Matrices, and Applications**, *University of Wyoming*
- Nov 2021 **Collaboration visit**, *MSRI (now SLMATH)*, Supervisor: L. Petrov, Project: "Asymptotics of noncolliding q -exchangeable random walks"

Service

- Sep 2025 - May 2026 **Directed Reading Program committee member**, *University of Virginia*
- Aug 2025 **Speaker at Summer MATH Teaching Workshop**, *University of Virginia*
- Jan - May 2025 **Directed Reading Program Mentor**, *University of Virginia*
- Mar, Sep 2025 **Manuscript Reviewer**, *Annales Henri Poincaré*

Apr 2025 **Manuscript Reviewer**, *Electronic Journal of Probability*
Sep 2024 **Manuscript Reviewer**, *Journal of Statistical Physics*
Jul 2024 **Organizer**, *Virginia Integrable Probability Summer School 2024*, <https://lpetrov.cc/vipss2024/>
Apr 2020 **Organizer**, *Online conference on Statistical Mechanics, Integrable Systems and Probability*, <http://mtikhonov.com/smisp/>

Teaching

F25, F23, S22 **Survey of Calculus I (MATH 1210)**, *Instructor*, University of Virginia, flipped classroom
Spring 2026 **Multivariable Calculus (MATH 2310)**, *Instructor*, University of Virginia
Spring 2025 **Calculus II (MATH 1320)**, *Instructor*, University of Virginia, flipped classroom
Su26, F22 **Calculus I (MATH 1310)**, *Instructor*, University of Virginia, flipped classroom
S21, F21 **Ordinary Differential Equations (MATH 3250)**, *TA, discussion sections*, University of Virginia
F20, Su21 **Multivariable Calculus (MATH 2310)**, *TA, discussion sections*, University of Virginia
F19, S20 **Symmetries and Group Theory**, *TA, discussion sections*, Lomonosov Moscow State University, flipped classroom
Fall 2019 **Relativity Theory for Pedestrians**, *Instructor and Course Author*, Kolmogorov Boarding School, Moscow State University, engagement course for high school students
S19, F19 **General Physics**, *Instructor*, Kolmogorov Boarding School, Moscow State University
Fall 2018 **Introduction to Differential Forms**, *TA, discussion sections*, Lomonosov Moscow State University

References

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